

POSTER PRESENTATION

IOT

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FUNCTIONAL BLOCKS OF TYPICAL SENSOR NODES IN IOT

These functional blocks are made up of sensors that handle interactions between a user device and the cloud, enable control and monitoring functions, manage data transfer, secure the IOT system through authentication, and various purposes, and offer an interface for monitoring and managing various concepts. All together make information about the IOT system available.

An important aspect in the development of a wireless sensor node is ensuring that there is always adequate energy available to power the system. The sensor node consumes power for sensing and communicating and data processing.

More energy is required for data communication than any other process. Power is stored either in batteries or capacitors. It uses both AC and DC power supply.

IOT systems are comprised of a number of building blocks including sensors, actuators, Radio, Power. The sensors and blocks are responsible for sensing, verification, actuation, management and communication.

The Sensor block serves as the data entry point in an IOT-system. Sensors collect data from their surroundings. Sensors gather data on temperature, humidity, light, motion, and other factors. These gadgets work together to collect data in the physical world.

- TEMPERATURE
- POSITION
- CHEMICAL
- FORCE
- LIGHT
- SPEED
- PRESSURE

ADC
SENSOR

PROCESSOR

ACTUATOR
INTERFACE

- PNEUMATIC
- HYDRAULIC
- MECHANICAL
- ELECTRICAL

RADIO

- WIFI
- BLUETOOTH
- LORA
- RFID/NFC
- ZIGBEE
- Z-WAVE
- GSM/3G/5G

Once the sensor/actuator block has collected data, it must be sent to the remainder of the system. This is when the connection block enters the picture. The radio block is a block which is in charge of creating and managing communication channels amongst IOT system devices. This can be accomplished with the use of several technologies such as Wi-Fi, Bluetooth, ZigBee and cellular networks.

The wireless capability of sensing the environment, they are set to measure and communicate the information to other sensor nodes or a remote host. Typically, a sensor node should have low-power requirements and be wireless. The wireless nature of sensor nodes would also allow them to be freely relocatable and deployed in large numbers without bothering about managing wires.

The Actuator block serves as the data entry point in an IOT system. Actuators drive physical processes.

Actuators turn on lights, open doors and control machines. These gadgets work together to operate in the physical world.

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